

Course Introduction

Programmable Networks A.Y. 2025/26

General Information

Programmable Networks (6 CFU)

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Office: room 109 (first floor), DIET Dept. (RM32, SPV)

Personal Zoom Room: <https://uniroma1.zoom.us/j/8967037141>

Office hour: appointment

Teaching material: Moodle, <https://elearning.uniroma1.it/course/view.php?id=19416>

Time schedule:

- Monday (13.30-15.00), Aula A6, DIAG
- Wednesday (13.30 - 16.00), Aula A4, DIAG

Exam

The subject is divided into two main parts:

- **theoretical lessons**
 - evaluated with an oral examination
 - 70% of the final grade
- **practical activities**
 - evaluated by developing a project
 - 30% of the final grade

The project is discussed and evaluated the same day of the exam

Objective

The main objective of this course is to let

- **Software engineers** to understand where/how they can contribute to Networking field
- **Network engineers** to understand what are the software development skills required to work in the Networking industry

This is achieved by

- introducing the **enabling technologies** for realizing a programmable network
- showing the most commonly used **APIs** and **programming languages**
- presenting the **reference architectures** to support a programmable network
- training the Network Applications development skills through “**hands on**” **labs**

PNet: Why?

- The Internet: A Remarkable Story
 - Tremendous success
 - From research experiment to global infrastructure
 - Brilliance of under-specifying
 - Network: best-effort packet delivery
 - Hosts: arbitrary applications
 - Enables innovation in applications
 - Web, P2P, VoIP, social networks, virtual worlds
 - But, change is easy only at the edge...

PNet: Why?

- Inside the network: a different story...
- Closed equipment
 - Software bundled with hardware
 - Vendor-specific interfaces
- Over specified
 - Slow protocol standardization
- Few people can innovate
 - Equipment vendors write the code
 - Long delays to introduce new features

Impacts performance, security, reliability, cost...

PNet: Why?

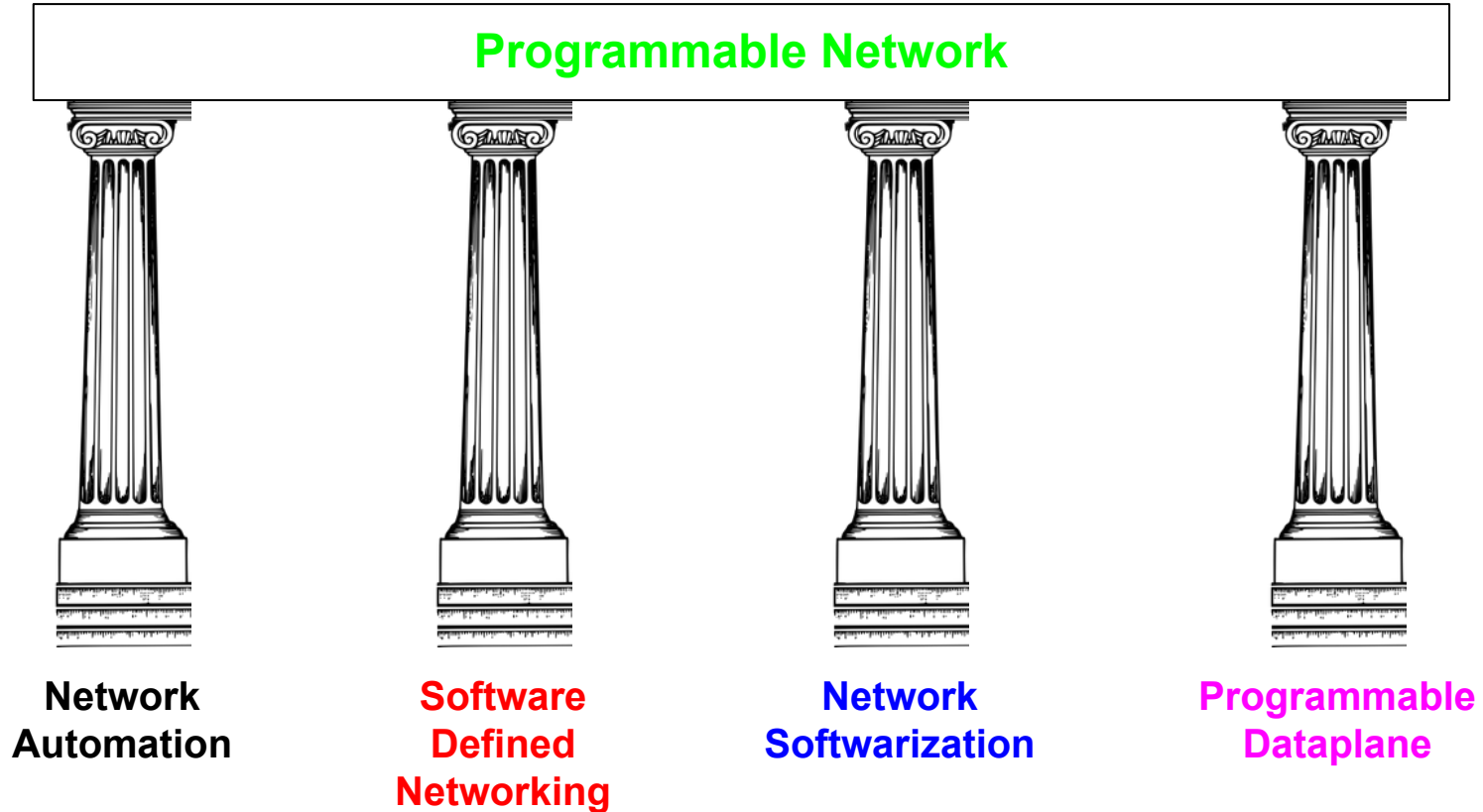
- Networks are **Hard to Manage**
 - Operating a network is expensive
 - More than half the cost of a network
 - Yet, operator error causes most outages
 - Buggy software in the equipment
 - Routers with 20+ million lines of code
 - Cascading failures, vulnerabilities, etc.
 - The network is “in the way”
 - Especially a problem in data centers with large numbers of VMs

PNet: Why?

os·si·fi·ca·tion  *noun* \ä-sə-fə-'kā-shən\
Definition of OSSIFICATION  +1  Like

- 1 a** : the natural process of bone formation
- b** : the hardening (as of muscular tissue) into a bony substance
- 2** : a mass or particle of **ossified** tissue
- 3** : a tendency toward or state of being molded into a rigid, conventional, sterile, or unimaginative condition

The main pillars of a Programmable Network



(Tentative) Program

Theoretical lessons on the following topics (30 hours)

- Networking basics
- Netconf/YANG
- SDN and Openflow
- NFV, NFV placement and Network Slicing
- SFC
- Segment Routing
- Programmable Data Plane
- P4 e P4Runtime

(Tentative) Program

Practical lessons on the following topics (30 hours)

- Netconf/YANG hands on
- SDN Lab
 - ONOS Tutorial
 - POX and Openflow
 - app development
- SRv6 configuration
- P4 Lab
 - P4 programming
 - P4runtime

